

A Renormalisation-Group Extension of Algebraic Connectivity Modeling in Quantum Networks

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Abstract—Quantum networks and computational architectures can be represented as weighted graphs in which vertices represent quantum resources, operations, memories, repeaters, or subsystems, while edges represent interaction or communication pathways. A central problem is the identification of weakly connected regions, or bottlenecks, which may influence information transport, entanglement distribution, coherence, and collective dynamical behaviour.

The Quantum Fiedler Eigenbasis for Rewiring Networks (QFERN) framework uses spectral graph theory to identify structurally important regions through the graph Laplacian, its Fiedler eigenvalue and eigenvector, and associated spectral sensitivities. Independently, the Kuramoto model provides a dynamical description of synchronization in networks of coupled oscillators. The Moore–Penrose pseudoinverse of the graph Laplacian provides a natural mathematical bridge between these descriptions because it simultaneously determines the linear network response to frequency forcing and the effective resistance between nodes.

This paper develops a proposed multiscale extension in which QFERN spectral information is used to construct a graph renormalisation transformation. The transformation coarse-grains nodes according to low-frequency Laplacian modes and produces a reduced Laplacian on the coarse variables. The Moore–Penrose pseudoinverse is then evaluated at each scale, allowing effective resistance and Kuramoto phase response to be tracked under coarse-graining. A six-node two-cluster graph is worked out analytically. The example demonstrates that the Fiedler mode captures the inter-cluster bottleneck, that the pseudoinverse amplifies this weak mode, and that a coarse-grained two-node graph retains the same nontrivial Fiedler eigenvalue.

The resulting framework suggests a scale-dependent workflow, $L^{(0)} \rightarrow (f_2^{(0)}, \lambda_2^{(0)}) \rightarrow P \rightarrow L^{(1)} \rightarrow (L^{(1)})^\dagger \rightarrow R^{(1)} \rightarrow \text{response}$, followed, where appropriate, by QFERN-guided rewiring. The construction is proposed as a bridge between spectral network optimisation and renormalisation-group ideas; it is not claimed that classical Kuramoto observables are themselves quantum-information observables. Establishing the mapping to quantities such as entanglement fidelity, coherence, or quantum-channel performance remains an open problem.

I. INTRODUCTION

Quantum networks may be represented by graphs in which nodes denote quantum resources and edges represent physical, logical, or information-transfer relationships. In a quantum repeater network, for example, nodes may represent memories or repeaters, while edges represent available entanglement resources. More generally, graph representations can be used to study computational architectures, routing structures, and networks of interacting quantum subsystems.

A recurring problem is the identification of bottlenecks. A network may contain many nodes and edges while still

possessing a weak large-scale connection between two regions. Spectral graph theory provides a natural way of detecting such structures through the eigenvalues and eigenvectors of the graph Laplacian [2].

The QFERN framework, introduced as the *Quantum Fiedler Eigenbasis for Rewiring Networks*, applies Fiedler-vector information and spectral derivatives to the analysis and rewiring of quantum resource graphs [1]. The framework considers quantum directed acyclic graphs (QDAGs) and examines how changes to graph structure affect spectral connectivity and effective-resistance-related quantities [1].

A second mathematical framework arises from synchronization theory. The Kuramoto model describes coupled oscillators whose phases evolve according to their natural frequencies and network couplings [4]. In the near-synchronous regime, the nonlinear sine coupling can be linearised, producing a Laplacian equation. This makes the Moore–Penrose pseudoinverse of the Laplacian a natural response operator [5], [6].

The same pseudoinverse determines effective resistance:

$$R_{ij} = (e_i - e_j)^T L^\dagger (e_i - e_j).$$

Consequently, the quantities

$$f_2, \quad \lambda_2, \quad L^\dagger, \quad R_{ij}$$

provide different but mathematically related descriptions of connectivity and network response.

A further connection arises from renormalisation-group (RG) theory. Sayama introduces RG analysis in the context of cellular automata and percolation by defining a coarse-scale connectivity variable and studying its transformation under changes of scale [3]. In that treatment, the central distinction is between dynamics across time and transformations across scale [3].

This observation motivates the following question:

Can the Fiedler eigenbasis and the Laplacian pseudoinverse be used to construct a scale-dependent renormalisation transformation for quantum-network graphs?

The purpose of this paper is not to claim that such a theory is already established. Instead, a concrete mathematical construction is proposed and tested on a tractable six-node network.

The proposed workflow is

spectral diagnosis $\rightarrow$ spectral coarse-graining $\rightarrow$ pseudoinverse response

The main contribution is therefore a bridge between QFERN-style spectral optimisation and RG-style scale transformations.

II. GRAPH LAPLACIAN

Consider a weighted undirected graph

$$G = (V, E, A),$$

where $A \in \mathbb{R}^{N \times N}$ is a symmetric weighted adjacency matrix.

The degree matrix is

$$D_{ii} = \sum_{j=1}^N A_{ij},$$

and the graph Laplacian is

$$L = D - A.$$

For a connected undirected graph,

$$0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N.$$

The zero eigenvalue follows from

$$L\mathbf{1} = 0.$$

The second-smallest eigenvalue,

$$\lambda_2,$$

is the algebraic connectivity. Its associated eigenvector f_2 is the Fiedler vector:

$$Lf_2 = \lambda_2 f_2.$$

The Fiedler eigenvector is orthogonal to the global phase mode:

$$f_2^T \mathbf{1} = 0.$$

The interpretation relevant to QFERN is that the Fiedler mode provides a spectral representation of a graph's weakest nontrivial connectivity mode. A small λ_2 is associated with weak global connectivity, while the components of f_2 identify the corresponding separation structure [1], [2].

III. THE MOORE–PENROSE PSEUDOINVERSE

Because

$$L\mathbf{1} = 0,$$

the Laplacian is singular and therefore has no ordinary inverse.

Let the orthonormal eigenvectors of L be

$$\{f_1, f_2, \dots, f_N\},$$

where

$$f_1 = \frac{1}{\sqrt{N}} \mathbf{1}.$$

The spectral decomposition is

$$L = \sum_{k=2}^N \lambda_k f_k f_k^T.$$

The Moore–Penrose pseudoinverse is

$$L^\dagger = \sum_{k=2}^N \frac{1}{\lambda_k} f_k f_k^T.$$

This expression reveals a direct relationship between weak Laplacian modes and network response.

In particular, the Fiedler contribution is

$$L_{(2)}^\dagger = \frac{1}{\lambda_2} f_2 f_2^T.$$

Thus, when

$$\lambda_2 \ll 1,$$

the corresponding Fiedler mode can make a large contribution to $L^\dagger$.

The three quantities therefore have complementary roles:

$$f_2 \rightarrow \text{spatial structure of weak connectivity,}$$

$$\lambda_2 \rightarrow \text{strength of the weak mode,}$$

and

$$L^\dagger \rightarrow \text{response to forcing in all nontrivial modes.}$$

IV. EFFECTIVE RESISTANCE

Let e_i denote the i -th standard basis vector. The effective resistance between nodes i and j is

$$R_{ij} = (e_i - e_j)^T L^\dagger (e_i - e_j).$$

Equivalently,

$$R_{ij} = L_{ii}^\dagger + L_{jj}^\dagger - 2L_{ij}^\dagger.$$

Using the spectral representation,

$$R_{ij} = \sum_{k=2}^N \frac{[f_k(i) - f_k(j)]^2}{\lambda_k}.$$

The Fiedler contribution is therefore

$$R_{ij}^{(2)} = \frac{[f_2(i) - f_2(j)]^2}{\lambda_2}.$$

Consequently, a pair of nodes may have a large Fiedler-mode contribution to effective resistance when their Fiedler components are strongly separated and λ_2 is small.

This makes effective resistance particularly useful as a bridge between structural and dynamical descriptions of network bottlenecks.

V. KURAMOTO NETWORK AND LINEAR RESPONSE

Consider the network Kuramoto model

$$\dot{\theta}_i = \omega_i + K \sum_{j=1}^N A_{ij} \sin(\theta_j - \theta_i).$$

In vector notation,

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_N)^T, \quad \boldsymbol{\omega} = (\omega_1, \dots, \omega_N)^T.$$

For sufficiently small phase differences,

$$\sin(\theta_j - \theta_i) \approx \theta_j - \theta_i,$$

giving

$$\dot{\boldsymbol{\theta}} \approx \boldsymbol{\omega} - KL\boldsymbol{\theta}.$$

The global phase is a null direction, so define the mean-free frequency vector

$$\boldsymbol{\omega}_\perp = \left(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T \right) \boldsymbol{\omega}.$$

At a stationary state,

$$KL\boldsymbol{\theta}_\perp = \boldsymbol{\omega}_\perp,$$

and hence

$$\boldsymbol{\theta}_\perp = \frac{1}{K} L^\dagger \boldsymbol{\omega}_\perp.$$

Thus $L^\dagger$ can be interpreted as a topology-dependent linear phase-response operator.

The important distinction is that this is a linearised Kuramoto result. It should not be interpreted as an exact solution of the original nonlinear dynamics for arbitrary phase differences.

Dörfler and Bullo established a broader set of Laplacian-based conditions for synchronization and phase cohesiveness in networked Kuramoto systems [5], [6].

VI. QFERN SPECTRAL REWIRING

The QFERN framework uses the Fiedler eigenbasis to identify structurally important network regions and to guide graph modification [1].

For a weighted edge (i, j) , the Laplacian contribution is

$$L^{(ij)} = w_{ij}(e_i - e_j)(e_i - e_j)^T.$$

Therefore,

$$\frac{\partial L}{\partial w_{ij}} = (e_i - e_j)(e_i - e_j)^T.$$

For a simple normalized Fiedler eigenvalue, standard eigenvalue perturbation theory gives

$$\frac{\partial \lambda_2}{\partial w_{ij}} = f_2^T \frac{\partial L}{\partial w_{ij}} f_2.$$

Consequently,

$$\frac{\partial \lambda_2}{\partial w_{ij}} = [f_2(i) - f_2(j)]^2.$$

This expression is particularly useful for a QFERN-inspired rewiring procedure because an edge connecting nodes with strongly different Fiedler components has high sensitivity with respect to λ_2 .

The proposed multiscale extension retains this spectral sensitivity but evaluates it at multiple scales.

VII. RENORMALISATION AS SPECTRAL COARSE-GRAINING

Sayama's RG treatment illustrates the central idea of studying how a connectivity quantity transforms between microscopic and macroscopic scales [3]. The transformation can be represented schematically as

$$p^{(0)} \rightarrow p^{(1)} \rightarrow p^{(2)} \rightarrow \dots$$

For a network, the analogous object can be the Laplacian itself:

$$L^{(0)} \rightarrow L^{(1)} \rightarrow L^{(2)} \rightarrow \dots$$

The crucial design question is how to construct the coarse variables.

Rather than imposing a purely geometric block transformation, the present proposal uses the low-frequency spectral structure of L . Let

$$P \in \mathbb{R}^{N \times M}$$

be a coarse-graining matrix whose columns span the retained large-scale subspace.

The reduced Laplacian is defined by

$$L_c = P^T L P,$$

when the columns of P are orthonormal.

If P is constructed from approximately constant vectors on spectral clusters, then L_c retains the connectivity between those clusters.

This gives the proposed transformation

$$\mathcal{R} : L^{(n)} \mapsto L^{(n+1)} = P_n^T L^{(n)} P_n.$$

The corresponding pseudoinverse is

$$(L^{(n+1)})^\dagger.$$

A scale-dependent response sequence can therefore be defined:

$$L^{(0)\dagger}, \quad L^{(1)\dagger}, \quad L^{(2)\dagger}, \dots$$

The resulting framework may be viewed as a spectral analogue of position-space coarse-graining.

VIII. SIX-NODE QFERN-RG EXAMPLE

A. Network Definition

Consider a six-node graph consisting of two complete three-node clusters connected by a single bridge:

$$\mathcal{C}_A = \{1, 2, 3\}, \quad \mathcal{C}_B = \{4, 5, 6\}.$$

The intra-cluster edges are

$$(1, 2), (1, 3), (2, 3)$$

and

$$(4, 5), (4, 6), (5, 6),$$

while the inter-cluster bridge is

$$(3, 4).$$

All edge weights are initially equal to one.

The topology is therefore

$$\begin{array}{ccc} 1 & - & 2 \\ \backslash & & / \\ 3 & & \end{array} \longleftrightarrow \begin{array}{ccc} & 4 & \\ / & & \backslash \\ 5 & - & 6 \end{array}.$$

The bridge (3, 4) is the only connection between the two three-node subgraphs.

The adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

The degree matrix is

$$D = \text{diag}(2, 2, 3, 3, 2, 2).$$

Hence

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ -1 & -1 & 3 & -1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

B. Laplacian Spectrum

The graph possesses the eigenvalue spectrum

$$\text{spec}(L) = \left\{ 0, \frac{2}{3}, 3, 3, 3, 4 \right\}.$$

Thus

$$\lambda_2 = \frac{2}{3}.$$

A normalized Fiedler vector is

$$f_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

It is immediately apparent that

$$f_2(i) < 0 \quad \text{for } i \in \mathcal{C}_A,$$

while

$$f_2(i) > 0 \quad \text{for } i \in \mathcal{C}_B.$$

The Fiedler mode therefore identifies the two clusters separated by the bridge.

The corresponding Fiedler-mode contribution to the pseudoinverse is

$$L_{(2)}^\dagger = \frac{1}{\lambda_2} f_2 f_2^T = \frac{3}{2} f_2 f_2^T.$$

Since

$$f_2 f_2^T = \frac{1}{6} \begin{pmatrix} J_3 & -J_3 \\ -J_3 & J_3 \end{pmatrix},$$

where J_3 is the 3×3 all-ones matrix, one obtains

$$L_{(2)}^\dagger = \frac{1}{4} \begin{pmatrix} J_3 & -J_3 \\ -J_3 & J_3 \end{pmatrix}.$$

C. Full Moore–Penrose Pseudoinverse

The remaining nontrivial modes have eigenvalues 3 and 4. Using the spectral decomposition,

$$L^\dagger = \frac{3}{2} f_2 f_2^T + \sum_{\lambda_k=3} \frac{1}{3} f_k f_k^T + \frac{1}{4} f_6 f_6^T.$$

For the present graph this gives the explicit matrix

$$L^\dagger = \begin{pmatrix} \frac{17}{36} & -\frac{19}{36} & -\frac{1}{36} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{19}{36} & \frac{17}{36} & -\frac{1}{36} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{36} & -\frac{1}{36} & \frac{17}{36} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{17}{36} & \frac{1}{36} & -\frac{19}{36} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{1}{36} & \frac{17}{36} & -\frac{19}{36} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & -\frac{19}{36} & -\frac{19}{36} & \frac{17}{36} \end{pmatrix}.$$

It satisfies

$$LL^\dagger = L^\dagger L = I - \frac{1}{6} J_6,$$

where $J_6 = \mathbf{1}\mathbf{1}^T$.

The matrix therefore inverts L on the six-dimensional space with the global phase direction removed.

D. Effective Resistances

For nodes in the same triangle,

$$R_{12} = R_{13} = R_{23} = \frac{2}{3},$$

and similarly

$$R_{45} = R_{46} = R_{56} = \frac{2}{3}.$$

For the bridge endpoints,

$$R_{34} = 1.$$

For a node on one side and a non-bridge node on the opposite side,

$$R_{14} = R_{24} = R_{35} = R_{36} = \frac{5}{3},$$

while the most distant cross-cluster pairs satisfy

$$R_{15} = R_{16} = R_{25} = R_{26} = \frac{7}{3}.$$

These values illustrate the distinction between a structural bridge and the complete pairwise resistance structure. The bridge itself has unit effective resistance, while cross-cluster pairs inherit additional resistance through their intra-cluster paths.

The Fiedler contribution to R_{ij} for any cross-cluster pair is

$$R_{ij}^{(2)} = \frac{\left(-\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}}\right)^2}{2/3} = 1.$$

Thus every cross-cluster pair contains a unit contribution from the weak inter-cluster mode.

IX. KURAMOTO RESPONSE ON THE SIX-NODE GRAPH

Choose a mean-zero frequency vector

$$\omega = \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

This vector is proportional to the Fiedler vector:

$$\omega = \sqrt{6} f_2.$$

Therefore,

$$L^\dagger \omega = \frac{1}{\lambda_2} \omega = \frac{3}{2} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

The linearised stationary phase response is consequently

$$\theta_\perp = \frac{3}{2K} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Thus

$$\theta_1 = \theta_2 = \theta_3 = -\frac{3}{2K},$$

and

$$\theta_4 = \theta_5 = \theta_6 = \frac{3}{2K}.$$

The phase difference across the inter-cluster bridge is therefore

$$\Delta\theta_{34} = \theta_4 - \theta_3 = \frac{3}{K}.$$

This is a particularly clear demonstration of the role of the pseudoinverse. The forcing vector is aligned exactly with the Fiedler mode, and therefore the response is amplified by

$$\frac{1}{\lambda_2} = \frac{3}{2}.$$

For the linearised model, the bridge carries the dominant phase separation because the natural-frequency mismatch is aligned with the weakest network mode.

For the original nonlinear Kuramoto model, the linearised solution is valid only while phase differences remain sufficiently small. A standard phase-cohesiveness analysis requires additional conditions involving K , the frequency vector, and the network topology [5], [6].

X. SPECTRAL COARSE-GRAINING

The Fiedler vector naturally partitions the graph into the two clusters

$$\mathcal{C}_A = \{1, 2, 3\}, \quad \mathcal{C}_B = \{4, 5, 6\}.$$

Define the orthonormal coarse-graining matrix

$$P = \begin{pmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 0 \\ 0 & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} \end{pmatrix}.$$

The columns of P represent uniform coarse variables on the two spectral clusters.

The reduced Laplacian is

$$L^{(1)} = P^T L P.$$

Direct evaluation gives

$$L^{(1)} = \begin{pmatrix} 1/3 & -1/3 \\ -1/3 & 1/3 \end{pmatrix}.$$

Its eigenvalues are

$$0, \quad \frac{2}{3}.$$

Hence

$$\lambda_2^{(1)} = \frac{2}{3} = \lambda_2^{(0)}.$$

The coarse-graining has therefore preserved the nontrivial Fiedler eigenvalue of this symmetric example.

The normalized coarse Fiedler vector is

$$f_2^{(1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The relationship between the microscopic and coarse Fiedler vectors is

$$f_2^{(0)} = P f_2^{(1)}.$$

This is the key result of the toy RG transformation: the Fiedler mode survives the coarse-graining exactly because the coarse variables were chosen from the spectral cluster structure.

A. Coarse Moore–Penrose Pseudoinverse

The pseudoinverse of the two-node coarse Laplacian is

$$(L^{(1)})^\dagger = \frac{3}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The coarse effective resistance is therefore

$$R_{AB}^{(1)} = (e_A - e_B)^T (L^{(1)})^\dagger (e_A - e_B),$$

giving

$$R_{AB}^{(1)} = 3.$$

This differs from the resistance of the microscopic bridge edge, which is 1, because the coarse nodes represent entire clusters. The coarse resistance therefore measures the effective response between the two aggregated variables rather than the resistance of a single physical edge.

This distinction is important for the proposed RG interpretation: coarse-grained observables need not equal individual microscopic observables. Instead, they encode the response of the collective degrees of freedom.

B. Coarse Kuramoto Response

Define the coarse frequency vector as

$$\omega^{(1)} = P^T \omega.$$

For the chosen frequency vector,

$$\omega^{(1)} = \sqrt{3} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The coarse stationary phase response is

$$\theta^{(1)} = \frac{1}{K} (L^{(1)})^\dagger \omega^{(1)}.$$

Hence

$$\theta^{(1)} = \frac{3\sqrt{3}}{2K} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Lifting this coarse solution back to the microscopic space gives

$$P\theta^{(1)} = \frac{3}{2K} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \end{pmatrix},$$

which is exactly the microscopic linearised solution.

Thus,

$$P(L^{(1)})^\dagger P^T \omega = L^\dagger \omega$$

for this particular forcing, because the forcing lies entirely in the retained Fiedler subspace.

This provides a concrete example of a scale-consistent response.

XI. THE QFERN–RG FLOW

The proposed renormalisation workflow can now be expressed as

$$L^{(0)} \xrightarrow{\text{spectral analysis}} \left(\lambda_2^{(0)}, f_2^{(0)} \right)$$

$$\left(\lambda_2^{(0)}, f_2^{(0)} \right) \xrightarrow{\text{clustering}} P_0$$

$$P_0^T L^{(0)} P_0 = L^{(1)}$$

$$L^{(1)} \longrightarrow (L^{(1)})^\dagger \longrightarrow R^{(1)} \longrightarrow \theta^{(1)}.$$

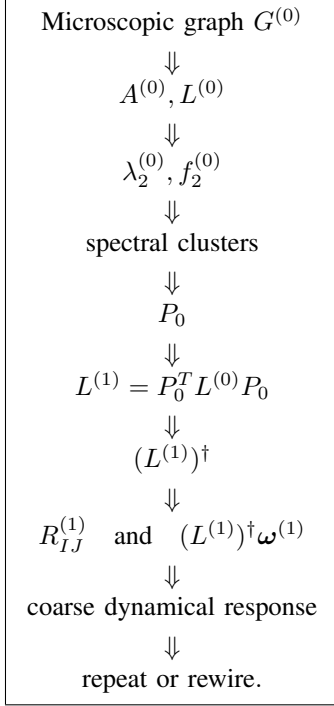
If another nontrivial scale exists, the process is repeated:

$$L^{(1)} \xrightarrow{\mathcal{R}} L^{(2)} \xrightarrow{\mathcal{R}} \dots$$

The complete proposed flow is therefore

TABLE I
PROPOSED QFERN–RG WORKFLOW.

Stage	Mathematical object	Function
1	A, L	Construct the network and its Laplacian.
2	λ_2, f_2	Identify the weakest nontrivial connectivity mode.
3	$\partial\lambda_2/\partial w_{ij}$	Measure Fiedler-based spectral sensitivity of edges.
4	P	Construct coarse variables from spectral clusters.
5	$L' = P^T L P$	Generate the coarse-scale network Laplacian.
6	$(L')^\dagger$	Compute coarse-scale network response.
7	R_{IJ}	Measure effective connectivity between coarse variables.
8	$(L')^\dagger \omega'$	Compute linearised coarse dynamical response.
9	$\mathcal{R}(L) = P^T L P$	Repeat the transformation to obtain an RG flow.
10	QFERN rewiring	Modify selected edges and recompute the spectral flow.



XII. UNIFIED QFERN–RG WORKFLOW

Table I summarises the proposed relationship without requiring each mathematical object to be interpreted independently.

The conceptual roles of the principal quantities are summarised in Table II.

XIII. MULTISCALE QFERN REWIRING

The previous construction allows QFERN to be interpreted as a scale-dependent optimisation procedure.

TABLE II
INTERPRETATION OF THE PRINCIPAL SPECTRAL AND DYNAMICAL QUANTITIES.

Quantity	Interpretation	Role in proposed framework
λ_2	Algebraic connectivity	Weak-mode strength
f_2	Fiedler eigenvector	Spectral partition
$L^\dagger$	Pseudoinverse	Network response
R_{ij}	Effective resistance	Pairwise connectivity
P	Coarse map	Scale transformation
L'	Reduced Laplacian	Coarse network
$B^T L^\dagger \omega$	Edge response	Phase-stress indicator
QFERN	Spectral rewiring	Network modification
RG	Scale transformation	Multiscale analysis

At scale n , define

$$L^{(n)} = D^{(n)} - A^{(n)}.$$

Compute

$$L^{(n)} f_2^{(n)} = \lambda_2^{(n)} f_2^{(n)}.$$

An edge sensitivity is then

$$s_{ij}^{(n)} = \left[f_2^{(n)}(i) - f_2^{(n)}(j) \right]^2.$$

Candidate edge modifications can be ranked according to this sensitivity subject to the physical constraints of the network.

After modification,

$$A^{(n)} \rightarrow \tilde{A}^{(n)},$$

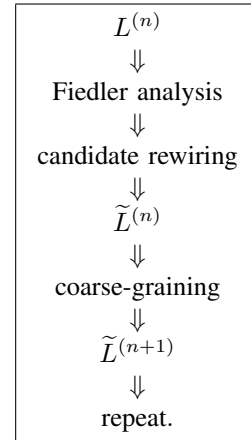
and therefore

$$L^{(n)} \rightarrow \tilde{L}^{(n)}.$$

The network is then coarse-grained again:

$$\tilde{L}^{(n+1)} = P_n^T \tilde{L}^{(n)} P_n.$$

This produces a coupled optimisation and renormalisation procedure:



The resulting object is not simply an optimised graph but a sequence

$$\{L^{(0)}, L^{(1)}, L^{(2)}, \dots\}$$

describing how the network's spectral structure evolves with scale.

XIV. POSSIBLE RG FIXED POINTS

The RG construction suggests a natural definition of a spectral fixed point.

Suppose that after coarse-graining and appropriate rescaling,

$$L^{(n+1)} \approx \alpha_n L^{(n)}.$$

A scale-invariant structure would satisfy

$$\mathcal{R}(L^*) \sim L^*.$$

The rescaling factor α_n is required because coarse-graining generally changes the numerical scale of the Laplacian.

A possible research programme is therefore to determine whether families of quantum-network topologies converge to common spectral fixed points.

For example, two microscopic networks

$$G_A^{(0)} \neq G_B^{(0)}$$

could nevertheless satisfy

$$\mathcal{R}^n(G_A^{(0)}) \rightarrow G_A^*,$$

and

$$\mathcal{R}^n(G_B^{(0)}) \rightarrow G_B^*,$$

with

$$G_A^* \sim G_B^*.$$

If so, the networks could belong to a common large-scale spectral class.

This is the point at which the proposed QFERN–RG construction becomes conceptually closer to renormalisation-group theory rather than merely hierarchical graph reduction.

XV. QUANTUM-NETWORK INTERPRETATION

The graph-theoretic construction above is mathematically classical. A graph Laplacian, its Fiedler vector, and its Moore–Penrose pseudoinverse do not by themselves define a quantum state or quantum channel.

Therefore, the proposed framework should be interpreted as a *network-level structural model* for a quantum architecture.

The mapping may be represented as

$$\text{quantum resources} \rightarrow \text{weighted graph} \rightarrow L \rightarrow L^\dagger \rightarrow \text{response}.$$

To establish a genuinely quantum RG theory, one would need an additional mapping

$$L^{(n)} \longrightarrow \mathcal{H}^{(n)} \quad \text{or} \quad \rho^{(n)},$$

where $\mathcal{H}^{(n)}$ denotes an effective Hamiltonian and $\rho^{(n)}$ an effective density operator.

The coarse-graining would then need to act on the quantum degrees of freedom:

$$\rho^{(n)} \rightarrow \rho^{(n+1)}.$$

Alternatively, one could associate the graph Laplacian with a single-excitation Hamiltonian,

$$H_{\text{graph}} = \hbar g L,$$

or with another physically justified graph-dependent operator.

The resulting theory would have to establish which quantum observables are preserved, transformed, or renormalised.

Potential observables include

$$\mathcal{F}_{\text{ent}}, \quad C(t), \quad \mathcal{C}_{\text{coh}}, \quad C_{\text{channel}},$$

representing, respectively, entanglement fidelity, coherence measures, coherence time, and channel capacity.

The present paper does not assert an equivalence between these quantities and effective resistance. Rather, it establishes the mathematical network-level bridge and identifies the additional physical mapping required.

XVI. DIRECTED QUANTUM NETWORKS AND THE UNDIRECTED SURROGATE

The QFERN framework considers Quantum Directed Acyclic Graphs in which directionality can encode the flow of quantum resources [1].

The six-node example in this paper is intentionally undirected. Consequently, its Laplacian is symmetric and positive semidefinite, which permits the standard Fiedler and effective-resistance construction.

For a genuinely directed graph,

$$A \neq A^T,$$

and the conventional symmetric Laplacian

$$L = D - A$$

may no longer possess all of the properties used above. Several possibilities then arise.

First, one may construct a symmetrised graph,

$$A_{\text{sym}} = \frac{1}{2}(A + A^T),$$

and analyse its undirected Laplacian.

Second, one may use a directed Laplacian or a Hermitian graph operator designed to preserve directional information.

Third, one may construct separate left and right spectral bases.

The appropriate choice depends on the physical meaning assigned to the edges.

Thus, the six-node construction should be regarded as a controlled proof-of-concept for the mathematical bridge rather than a complete treatment of directed quantum networks.

XVII. LIMITATIONS AND OPEN PROBLEMS

Several issues must be resolved before the proposed construction can be considered a general quantum-network renormalisation theory.

A. Choice of Coarse Variables

The matrix P is not unique. The present example chooses P from the Fiedler partition, but a general network may require several low-frequency eigenvectors,

$$P = [f_1 \ f_2 \ \cdots \ f_m],$$

or a clustering constructed from them.

A principled criterion for selecting the retained spectral modes is therefore required.

B. Directed Graphs

Quantum communication networks may be directional. The relationship between QFERN's QDAG representation and symmetric effective-resistance theory requires further development.

C. Nonlinear Kuramoto Dynamics

The relation

$$\theta = \frac{1}{K} L^\dagger \omega$$

is a linearised result. The full nonlinear dynamics are governed by

$$\dot{\theta}_i = \omega_i + K \sum_j A_{ij} \sin(\theta_j - \theta_i).$$

A full theory must therefore track the nonlinear phase differences under coarse-graining.

D. Quantum Observables

A graph-theoretic resistance is not automatically a quantum information metric. The relationship between effective resistance and quantities such as entanglement fidelity, concurrence, entropy, coherence, or quantum-channel capacity must be derived for each physical architecture.

E. RG Fixed Points

A meaningful RG theory requires a well-defined transformation, rescaling prescription, and notion of fixed point. The present construction establishes a candidate transformation

$$L^{(n+1)} = P_n^T L^{(n)} P_n,$$

but does not yet establish universality classes or critical exponents for quantum networks.

XVIII. PROPOSED RESEARCH PROGRAMME

The mathematical construction suggests a sequence of increasingly strong tests.

- 1) Construct families of weighted quantum-network graphs with controllable bottlenecks.
- 2) Compute

$$\lambda_2, \ f_2, \ L^\dagger, \ R_{ij}$$

for each graph.

- 3) Construct spectral coarse-graining matrices P .
- 4) Compare microscopic and coarse-scale pseudoinverse responses.
- 5) Apply the construction to Kuramoto dynamics and measure phase cohesiveness.
- 6) Apply QFERN-style rewiring at each scale.
- 7) Determine whether the resulting networks exhibit scale-invariant spectral structures.
- 8) Introduce an explicit quantum model and test whether graph-level RG observables correlate with quantum-information observables.
- 9) Search for fixed points and universality classes.

A useful numerical experiment would therefore measure the vector

$$\mathcal{S}^{(n)} = \left(\lambda_2^{(n)}, R_{\max}^{(n)}, \|L^{(n)\dagger} \omega^{(n)}\|, \Phi^{(n)} \right),$$

where $R_{\max}$ denotes a suitable maximum or percentile effective resistance and Φ denotes a dynamical phase-cohesiveness observable.

The RG flow would then be represented by

$$\mathcal{S}^{(0)} \rightarrow \mathcal{S}^{(1)} \rightarrow \mathcal{S}^{(2)} \rightarrow \dots$$

XIX. RESULTS

A. QFERN Network Construction and Spectral Coarse-Graining

The QFERN-renormalisation-group (RG) procedure was evaluated using clustered networks with user-defined numbers of clusters, nodes per cluster, and intra-cluster topology. The implementation supports Watts–Strogatz, Barabási–Albert, karate-club, star, ring, and line networks. For a microscopic graph $G^{(0)}$, the weighted Laplacian is

$$L^{(0)} = D - A, \tag{1}$$

where A is the weighted adjacency matrix and D is the degree matrix. Its eigendecomposition is

$$L^{(0)} V = V \Lambda. \tag{2}$$

The second eigenvalue $\lambda_2^{(0)}$ and corresponding eigenvector $\mathbf{v}_2^{(0)}$ define the Fiedler mode used for the spectral partition.

The microscopic network and its Fiedler structure are shown in Figs. 1 and 2.

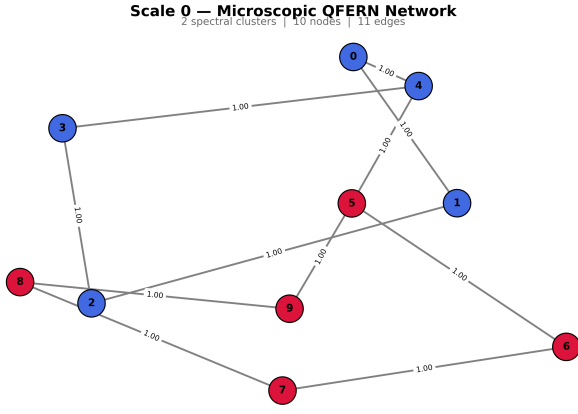


Fig. 1. Microscopic QFERN network at scale 0. Highlighted nodes identify the transition region associated with the spectral partition.

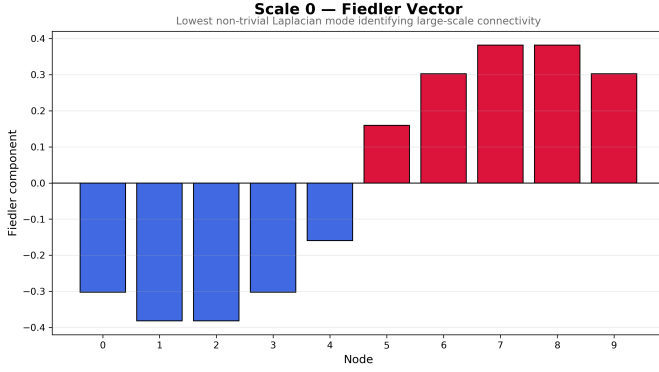


Fig. 2. Microscopic Fiedler vector $\mathbf{v}_2^{(0)}$, showing the spectral separation of the microscopic nodes.

B. Laplacian Spectrum, Effective Resistance and QFERN Sensitivity

The Laplacian eigenspectrum is shown in Fig. 3. The zero eigenvalue represents the uniform mode, while $\lambda_2^{(0)}$ defines the algebraic connectivity and its associated Fiedler mode.

The Moore–Penrose pseudoinverse of the Laplacian is

$$L^\dagger = \sum_{\lambda_k > 0} \frac{1}{\lambda_k} \mathbf{v}_k \mathbf{v}_k^T. \quad (3)$$

The effective resistance between nodes i and j is then

$$R_{ij} = (\mathbf{e}_i - \mathbf{e}_j)^T L^\dagger (\mathbf{e}_i - \mathbf{e}_j). \quad (4)$$

The resulting resistance matrix is shown in Fig. 14.

The QFERN edge sensitivity follows from the Fiedler mode as

$$\frac{\partial \lambda_2}{\partial w_{ij}} (v_{2,i} - v_{2,j})^2. \quad (5)$$

Edges joining nodes with larger Fiedler-component differences therefore have greater sensitivity to changes in their

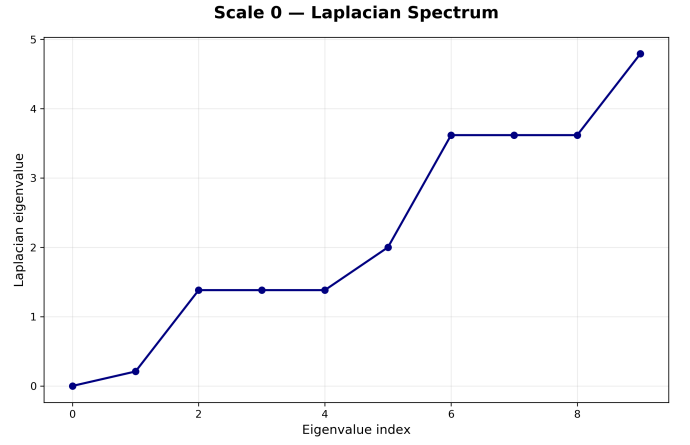


Fig. 3. Laplacian eigenspectrum of the microscopic QFERN network. The second eigenvalue $\lambda_2^{(0)}$ defines the Fiedler mode used for spectral partitioning.

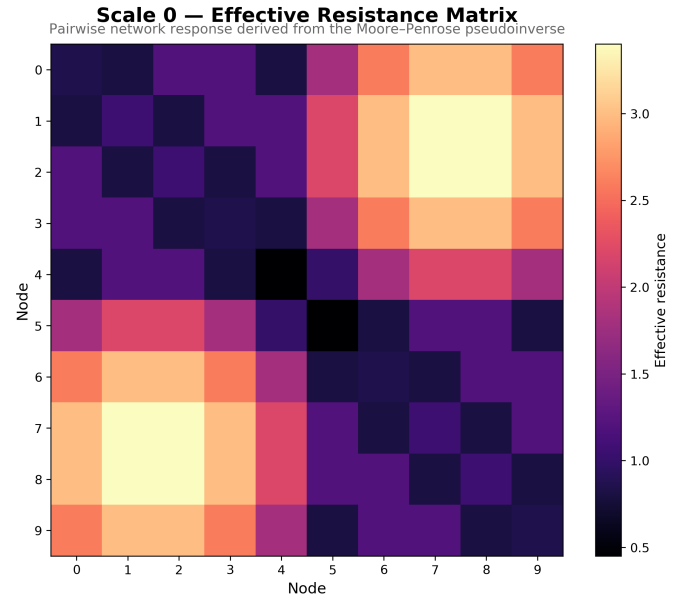


Fig. 4. Effective-resistance matrix of the microscopic QFERN network.

coupling weights. The resulting edge sensitivities are shown in Fig. 5.

C. Construction of the Coarse-Grained Network

The microscopic node set is partitioned into M clusters,

$$\mathcal{C} = C_1, C_2, \dots, C_M. \quad (6)$$

The normalized coarse-graining matrix is defined by

$$P_{ia} = \begin{cases} |C_a|^{-1/2}, & i \in C_a \\ 0, & i \notin C_a. \end{cases} \quad (7)$$

For non-overlapping clusters,

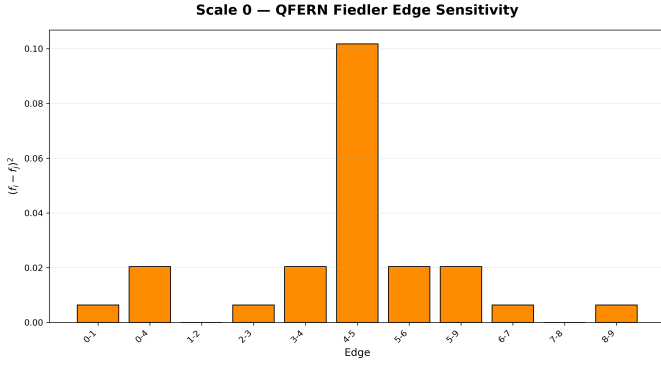


Fig. 5. QFERN edge sensitivity $\partial\lambda_2/\partial w_{ij}$ derived from the microscopic Fiedler vector.

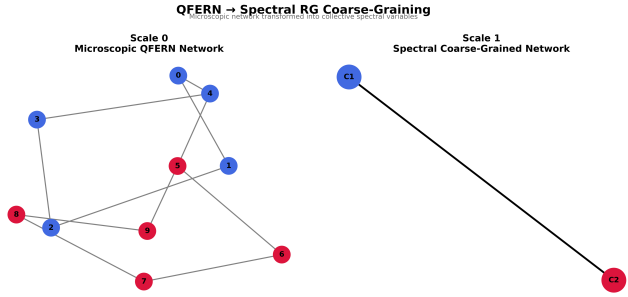


Fig. 6. QFERN spectral RG flow from microscopic scale 0 to coarse scale 1.

$$P^T P = I_M. \quad (8)$$

The coarse Laplacian is

$$L^{(1)} = P^T L^{(0)} P. \quad (9)$$

Each coarse node therefore represents a collective cluster variable. The resulting scale-1 network is shown in Fig. 6.

D. Microscopic-to-Coarse RG Flow

The complete transformation is illustrated in Fig. 6. The microscopic graph is mapped to collective cluster variables according to

$$G^{(0)} \rightarrow L^{(0)} \rightarrow \mathbf{v}_2^{(0)} \rightarrow P \rightarrow L^{(1)} \rightarrow G^{(1)}. \quad (10)$$

E. Four-Cluster QFERN Network and Spectral RG

To further demonstrate the utility of the QFERN spectral renormalisation-group construction, we consider a more complex four-cluster network consisting of four identical five-node ring subnetworks connected by a chain of inter-cluster bridges. The microscopic network therefore contains

$$N = 20 \quad (11)$$

QFERN + SPECTRAL RG Simulation

Clusters 2
Topologies Ring, Ring
Bridge Chain
Nodes 10
Edges 11

Clusters:

$$C_1 = \{0, 1, 2, 3, 4\}, \quad C_2 = \{5, 6, 7, 8, 9\}$$

Eigenvalues:

$$\lambda^{(0)} = \{0, 0.208712, 1.381966, 1.381966, 1.381966, 2, 3.618034, 3.618034, 3.618034, 3.618034\}$$

$$\lambda^{(1)} = \{0, 0.400000\}$$

$$\lambda_2^{(0)} = 0.2087121525$$

$$\lambda_2^{(1)} = 0.4000000000$$

$$\epsilon_F = 2.5930 \times 10^{-1}$$

$$\epsilon_\theta = 2.5339 \times 10^0$$

Fig. 7. Numerical summary of the QFERN spectral RG simulation.

nodes and

$$E = 23 \quad (12)$$

edges. The microscopic node set is partitioned into four clusters,

$$\mathcal{C} = \{C_1, C_2, C_3, C_4\}, \quad (13)$$

with

$$C_1 = \{0, 1, 2, 3, 4\},$$

$$C_2 = \{5, 6, 7, 8, 9\},$$

$$C_3 = \{10, 11, 12, 13, 14\},$$

$$C_4 = \{15, 16, 17, 18, 19\}. \quad (14)$$

Each cluster has ring topology, while the clusters are connected according to a chain topology. This construction provides a useful test case because the microscopic network contains both local intra-cluster modes and lower-frequency collective modes associated with motion between clusters.

The resulting microscopic network and its corresponding four-cluster coarse representation are shown in Fig. 8.

The scale-0 Laplacian is again defined by

$$L^{(0)} = D - A, \quad (15)$$

and its eigendecomposition is

$$L^{(0)} V^{(0)} = V^{(0)} \Lambda^{(0)}. \quad (16)$$

For this four-cluster network, the complete microscopic eigenvalue spectrum generates a corresponding spectrum shown in Fig. 9. In particular, the first four eigenvalues form a low-lying spectral sector,

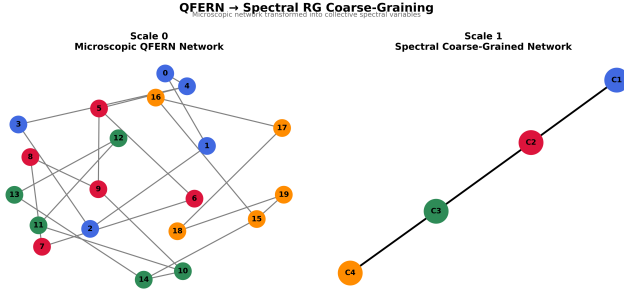


Fig. 8. Four-cluster QFERN network used for the spectral RG demonstration. The scale-0 network contains four five-node ring clusters connected by a chain of inter-cluster links. The scale-1 network contains four collective cluster variables, connected as a coarse chain.

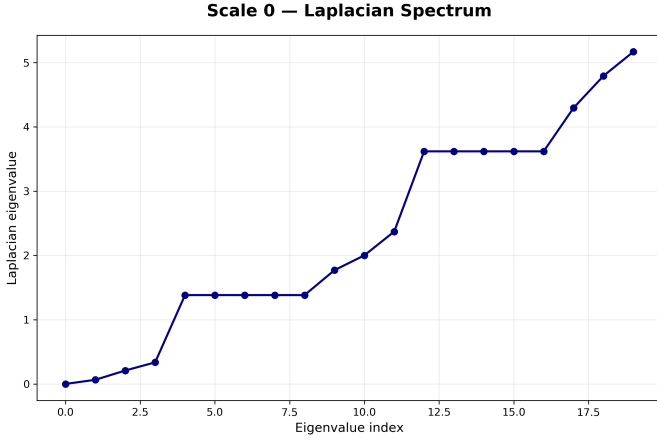


Fig. 9. Laplacian eigenspectrum of the 20-node four-cluster QFERN network. The low-lying modes occur below the intra-cluster spectral sector beginning near $\lambda \simeq 1.382$.

$$\lambda_{\text{low}}^{(0)} = \{0, 0.06408113, 0.20871215, 0.33542585\}, \quad (17)$$

followed by a substantial spectral separation before the repeated intra-cluster ring eigenvalues at

$$\lambda \simeq 1.381966. \quad (18)$$

This separation is significant for the RG construction: the low spectral modes describe collective variations between the four clusters, whereas the higher modes predominantly describe fluctuations internal to the individual ring subnetworks.

F. Four-Cluster Fiedler Mode and Edge Sensitivity

The second microscopic eigenvalue is

$$\lambda_2^{(0)} = 0.0640811308, \quad (19)$$

with associated Fiedler vector $\mathbf{v}_2^{(0)}$. Since the Fiedler vector represents the lowest non-trivial Laplacian mode, its components vary primarily across the large-scale cluster structure rather than across the individual nodes within each ring.

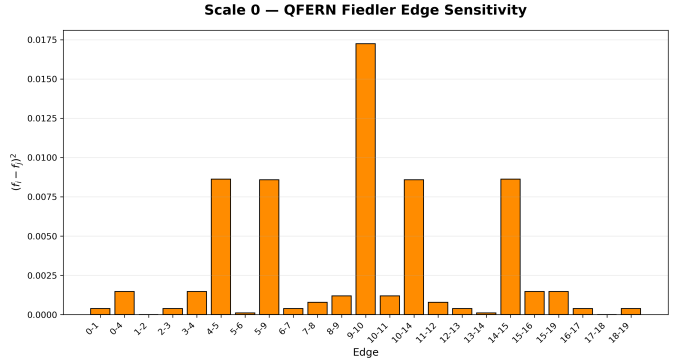


Fig. 10. QFERN Fiedler-edge sensitivity for the four-cluster microscopic network. The sensitivity is determined by the squared difference of the Fiedler-vector components across each edge.

The Fiedler mode therefore provides a microscopic description of the weakest collective direction of the network. This is precisely the mode used by the QFERN construction to identify spectrally important edges and to construct the corresponding coarse variables.

The sensitivity of the Fiedler eigenvalue to an edge weight is given by

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (v_{2,i} - v_{2,j})^2. \quad (20)$$

Consequently, edges whose endpoints have substantially different Fiedler components contribute more strongly to the sensitivity of the algebraic connectivity. In the four-cluster example, this provides a direct microscopic diagnostic for identifying edges that participate in the collective separation of the cluster structure.

The corresponding linearised Kuramoto phase response provides a complementary representation of the same spectral structure. The response is negative on one side of the network and positive on the other, with the sign change occurring across the collective transition region. This demonstrates how the spectral mode can also be interpreted dynamically as a collective phase-response pattern.

G. Spectral Coarse-Graining of the Four-Cluster Network

The microscopic node variables are projected onto the four cluster variables using the normalized coarse-graining matrix

$$P_{ia} = \begin{cases} |C_a|^{-1/2}, & i \in C_a, \\ 0, & i \notin C_a. \end{cases} \quad (21)$$

Because the four clusters are non-overlapping,

$$P^T P = I_4. \quad (22)$$

The scale-1 Laplacian is then obtained by projection,

$$L^{(1)} = P^T L^{(0)} P. \quad (23)$$

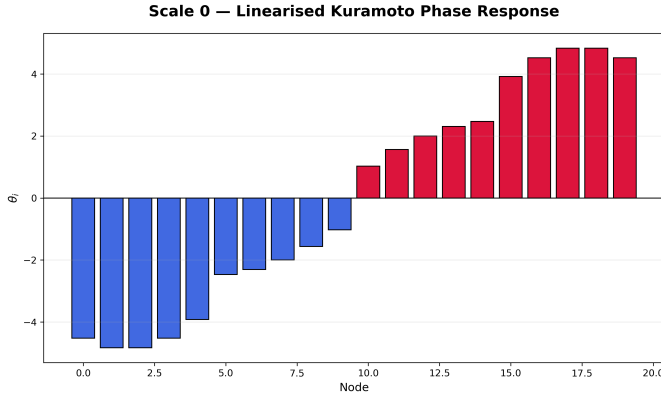


Fig. 11. Linearised Kuramoto phase response associated with the four-cluster QFERN network. The sign structure separates the collective response of the lower and upper portions of the microscopic network.

For the present network, the resulting coarse-grained spectrum is

$$\lambda^{(1)} = \{0, 0.11715729, 0.40000000, 0.68284271\}. \quad (24)$$

Thus, the original 20-node spectral problem is reduced to a four-dimensional collective problem. Importantly, the number of coarse variables is identical to the number of structurally defined clusters,

$$20 \text{ microscopic nodes} \longrightarrow 4 \text{ collective variables}. \quad (25)$$

The scale-1 eigenvalues retain the ordered low-frequency structure of the microscopic network while removing the higher-frequency intra-cluster modes. The coarse spectrum therefore provides a compact representation of the collective network dynamics.

The microscopic and coarse Fiedler eigenvalues are

$$\lambda_2^{(0)} = 0.0640811308, \quad (26)$$

and

$$\lambda_2^{(1)} = 0.1171572875, \quad (27)$$

respectively. The difference between the two values reflects the change in spectral scale associated with replacing the microscopic degrees of freedom by normalized collective cluster variables.

H. Fiedler and Response Lifting Across Scales

The quality of the coarse representation can be assessed by lifting the scale-1 collective modes back to the microscopic space. Given a coarse eigenvector $\mathbf{u}^{(1)}$, its microscopic reconstruction is

$$\tilde{\mathbf{u}}^{(0)} = P\mathbf{u}^{(1)}. \quad (28)$$

For the Fiedler mode, the normalized lifting error obtained for the four-cluster network is

$$\epsilon_F = 1.2615409768 \times 10^{-1}. \quad (29)$$

This provides a quantitative measure of the difference between the microscopic Fiedler structure and its four-variable coarse representation.

The corresponding response lifting error is

$$\epsilon_\theta = 7.2834314549. \quad (30)$$

The two errors quantify different aspects of the coarse-graining: ϵ_F measures the reconstruction of the spectral Fiedler mode, whereas ϵ_θ measures the reconstruction of the associated phase-response quantity. They should therefore be interpreted as separate diagnostics rather than as interchangeable measures of coarse-graining accuracy.

I. Utility of the Spectral RG Representation

The four-cluster example illustrates the principal utility of the QFERN spectral RG construction. At the microscopic level, the network contains 20 node variables and 23 edges, together with a large number of Laplacian modes describing both collective and intra-cluster fluctuations. The RG transformation replaces these microscopic variables with four normalized collective cluster variables,

$$G^{(0)} \longrightarrow L^{(0)} \longrightarrow \mathbf{v}_2^{(0)} \longrightarrow P \longrightarrow L^{(1)} \longrightarrow G^{(1)}. \quad (31)$$

The spectral structure provides a natural justification for this reduction. The first non-zero microscopic eigenvalue occurs at

$$\lambda_2^{(0)} \approx 0.0641, \quad (32)$$

while the next two collective modes occur at approximately

$$0.2087 \quad \text{and} \quad 0.3354. \quad (33)$$

These four low-lying modes are separated from the repeated intra-cluster eigenvalue near

$$\lambda \simeq 1.382. \quad (34)$$

The resulting spectral separation indicates that the lowest part of the spectrum is associated with collective inter-cluster structure, whereas the higher part contains predominantly local intra-cluster fluctuations. The RG transformation consequently retains the collective sector while integrating out the detailed internal degrees of freedom of each five-node ring.

This provides a useful interpretation of QFERN as a spectral reduction rather than simply a graph-size reduction. The coarse network is not obtained by arbitrarily deleting microscopic nodes; instead, each coarse node represents a collective subspace selected by the network's spectral structure. In this sense, the four-cluster example demonstrates how Fiedler-based analysis can connect microscopic edge sensitivity, collective phase response, and coarse-grained network dynamics within a single RG framework.

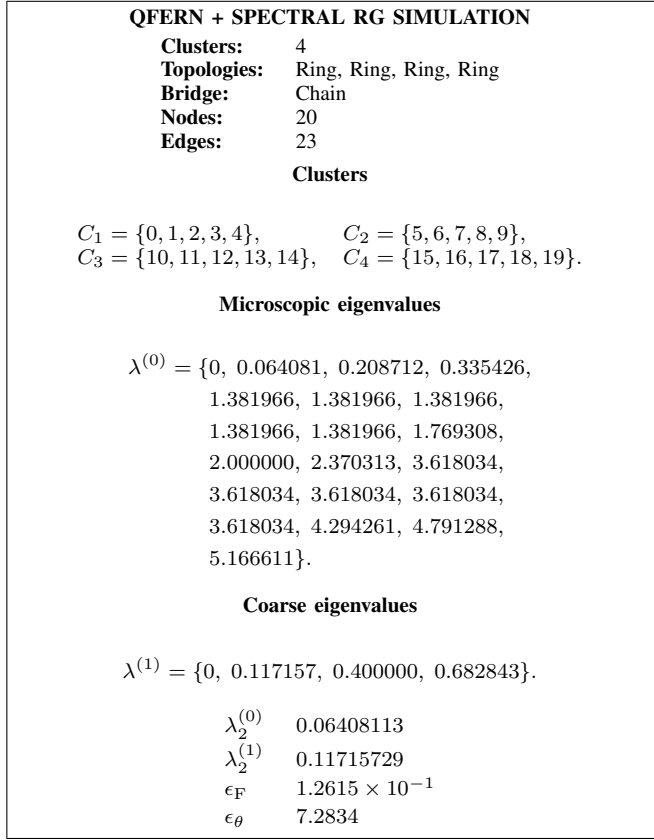


Fig. 12. Numerical summary of the four-cluster QFERN spectral RG simulation.

As seen, the effective-resistance matrix grows in complexity as the network and links themselves grow:

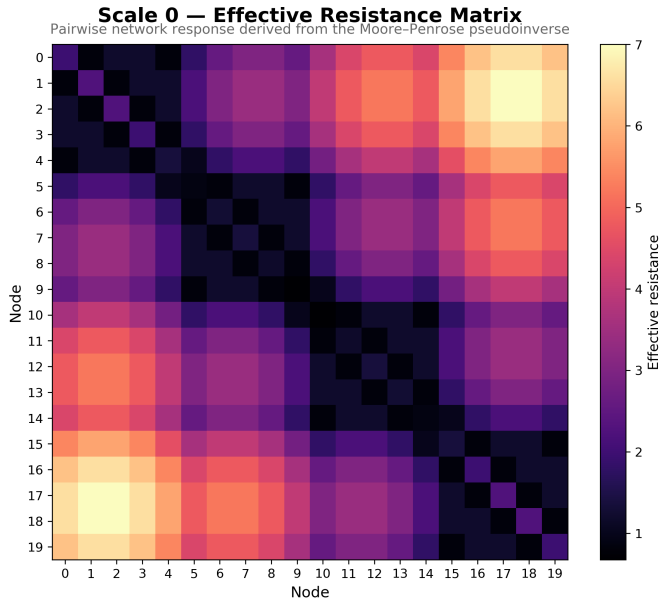


Fig. 13. Effective-resistance matrix of the microscopic QFERN network of four clusters.

So coarse graining is an intuitive solution to network topology in this regard. It even highlights areas of effective resistance more effectively, rounding them up to nearest positive integer value rather than washing them out as they are on the microscopic scale.

Scale 1 — Renormalised Effective Resistance Matrix

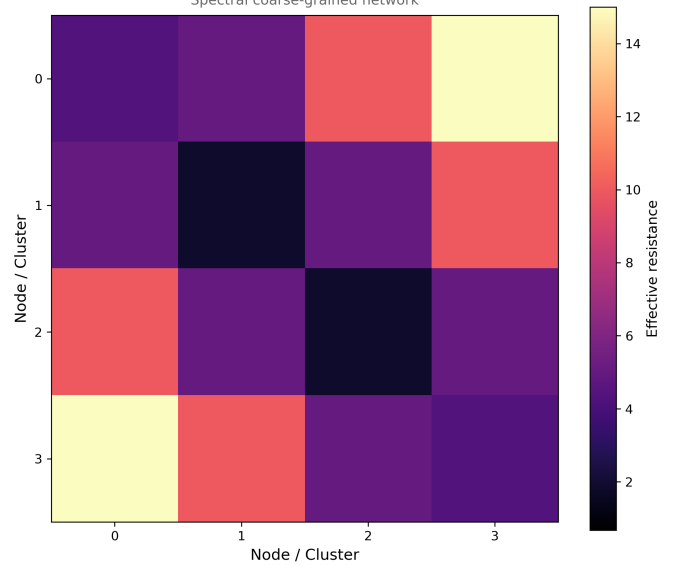


Fig. 14. Renormalised Effective-resistance matrix of the microscopic QFERN network of 4 clusters.

The extensions to QFERN covered here provides a common numerical framework for examining how network topology affects spectral coarse-graining while retaining the same underlying RG transformation.

XX. CONCLUSION

This paper has developed a proposed mathematical bridge between QFERN, the Moore–Penrose pseudoinverse, Kuramoto synchronization, and renormalisation-group ideas.

The fundamental spectral relationships are

$$L f_2 = \lambda_2 f_2,$$

$$L^\dagger = \sum_{k=2}^N \frac{1}{\lambda_k} f_k f_k^T,$$

and

$$R_{ij} = (e_i - e_j)^T L^\dagger (e_i - e_j).$$

For linearised Kuramoto dynamics,

$$\theta_\perp = \frac{1}{K} L^\dagger \omega_\perp.$$

These equations show that the Fiedler eigenvector identifies a weak connectivity mode, its eigenvalue determines the strength of that mode, the pseudoinverse determines network

response, and effective resistance measures pairwise connectivity.

The six-node example demonstrates a particularly transparent case. Two three-node clusters are connected by a single bridge, producing

$$\lambda_2 = \frac{2}{3}$$

and

$$f_2 = \frac{1}{\sqrt{6}}(-1, -1, -1, 1, 1, 1)^T.$$

The spectral coarse-graining matrix

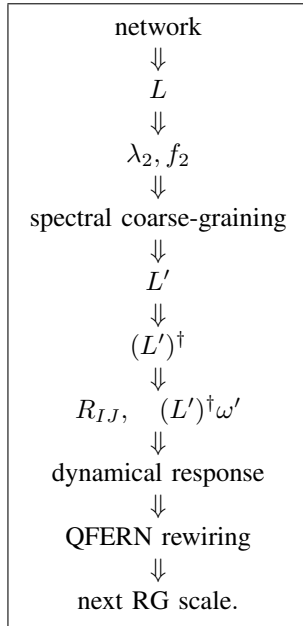
$$P = \begin{pmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 0 \\ 0 & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} \end{pmatrix}$$

produces

$$L^{(1)} = P^T L P = \begin{pmatrix} 1/3 & -1/3 \\ -1/3 & 1/3 \end{pmatrix},$$

whose nontrivial eigenvalue remains $2/3$. The corresponding coarse-grained Fiedler mode therefore preserves the dominant large-scale separation of the microscopic graph.

The proposed RG extension can be summarised as:



The important conceptual point is that QFERN can supply the *spectral diagnostic and control mechanism*, while RG supplies the *scale transformation*. The Moore–Penrose pseudoinverse then provides a common response operator connecting structural connectivity, effective resistance, and linearised oscillator dynamics.

The framework remains a proposed extension. In particular, the mapping from classical graph quantities and Kuramoto response to genuinely quantum observables has not been established here. A complete quantum formulation will require an explicit physical mapping from graph coarse-graining to quantum states, channels, Hamiltonians, or quantum resource measures.

Nevertheless, the six-node construction demonstrates that a mathematically consistent QFERN–RG workflow can be formulated and that, for a spectrally aligned forcing, the microscopic and coarse-grained pseudoinverse responses can agree exactly. This provides a concrete starting point for investigating multiscale spectral optimisation in quantum networks.

XXI. CODE AVAILABILITY

The code used to generate the numerical results and plots presented in this work is publicly available on GitHub:

[QFERNRGCoarseGrainV7.2.py](#)

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